

WEBSITE_DESIGN_PROPOSAL

Helix OTA — Public Marketing Website: Design Proposal

Revision: 1 **Last modified:** 2026-07-11T20:27:30Z **Status:** DESIGN PROPOSAL — analysis + design ONLY. No scaffold, no source, no build, no git run is performed under this document. This is the §11.4.167 first-artifact / brainstorming deliverable for the P4 marketing-website work-stream; per the brainstorming HARD-GATE the operator approves this design BEFORE any build, and the load-bearing choices are surfaced as decisions (§11.4.66), never silently taken. **Authority:** §11.4.190 (website engineering-quality mandate) · §11.4.162 (OpenDesign tokens) · §11.4.170 (device-independent host-rendered pixel proof) · §11.4.6 (no-guessing — every claim cites a path/source; every unknown marked UNCONFIRMED:). **Scope owner / track:** Helix OTA · T4 / **feature/website** · new submodule submodules/website (own .git, §11.4.179).

Relationship to prior artifacts: this proposal CONSOLIDATES and re-frames the two existing research artifacts into the requested design-proposal shape; it does not restate them byte-for-byte. Read alongside: docs/research/website/00_WEBSITE_DESIGN_AND_BUILD_PLAN.md (Rev 2 — the comprehensive design+build plan) and docs/research/website/01_SCAFFOLD_READINESS_AND_VERIFICATION.md (Rev 2 — the scaffold-time verification), and the production plan docs/planning/PRODUCTION_READINESS_PLAN.md §2.2 / §5 P4 / K13.

0. Grounded facts this proposal is built on (§11.4.6 — verified this session)

Fact	Source (read this session)
The marketing website does not exist — no website/site/www/	

Fact	Source (read this session)
dir, no website entry in .gitmodules, submodules/website absent. Firebase hosts only the two admin SPAs (dashboard, ota-manager), both meant to stay noindex. Only planning exists: docs/research/website/00_* + 01_* (Rev 2). Content is operator-LOCKED: sales email contact@hxota.com, NO pricing/packages/tiers anywhere (a “Contact sales” CTA in their place), footer “<i>Made with ♥ by Helix Development team</i>” (heart icon, not the word “love”). OpenDesign token package present at design-systems/helix-ota/ — tokens.css, tailwind-v4.css, DESIGN.md (+ .html/.pdf exports), manifest.json (schemaVersion: od-design-system-project/v1, id: helix-ota, category Enterprise). The shipped OpenDesign package is dashboard-first, ships a BLUE accent (--accent:#2563EB light / #3B82F6 dark), and its own DESIGN.md says it is “built for operators watching fleets ... not for marketing pages.” → the marketing site needs a Helix-green brand LAYER extending these tokens (designed in research §3), not the stock blue package as-is. Brand logo assets/Logo.png present (213 533 bytes). Brand greens: lime “Helix green” #B5E215—#BBE639, mint #B8ECD7.	ls of repo root + .gitmodules grep + firebase.json (two hosting targets only) docs/research/website/00_*.md §1.4 Read design-systems/helix-ota/* design-systems/helix-ota/ DESIGN.md L9-19,32 ls assets/Logo.png + research §0

Fact	Source (read this session)
Real system to describe honestly: Go/Gin control plane github.com/HelixDevelopment/helix_ota/ server; six Apache-2.0 ota- bricks; native Android A/B (update_engine + AVB/dm-verity + auto-rollback); targets RK3588 / Orange Pi 5 Max; Linux/Windows on the roadmap. Multi-tenant + real object storage are ROADMAP (planning-only).	PRODUCTION_READINESS_PLAN.md §1, §2.1, §2.2; research §0/§1.3
Angular 22 verified latest stable (patch v22.0.6 as of 2026-07-10); --ssr/--standalone(default)/-- package-manager=pnpm valid; Tailwind v4 @tailwindcss/ postcss + .postcssrc.json wiring matches the official guide; @jsverse/transloco v8 (signals) current; the three brand fonts are OFL 1.1 (self-host-clear).	research 01_* §2 (each with a fetched source)

Non-fiction rule (anti-bluff at the marketing layer, §11.4). Every user-visible claim on the site MUST map to a shipping capability. Roadmap items (multi-tenant, real object storage, Linux/Windows targets, the upload CLI) MUST be visually marked “Coming / On the roadmap” — a marketing site that claims an unshipped feature works is the same defect class as a PASS-bluff.

1. Page / section inventory (proposed)

Recommendation: a single long-scroll home (/) as the launch surface, structured as nine narrative sections, plus two optional deep routes deferred until content depth justifies them, plus the required machine files. This maximizes SEO focus and Core Web Vitals for launch while keeping deep content linkable later. (Synthesized from research §1.1 narrative arc + §1.2 sitemap.)

1.1 Launch surface — / (nine sections, one indexable route)

#	Section	Job (message intent — final copy is a later, operator-owned step)	Maturity shown
S0	Masthead / Hero	One-line promise (“Ship firmware to your whole fleet. Never brick a device.”), animated “Living Helix” motif, mono version-string ticker, one green CTA + one ghost CTA.	—
S1	Simple story	Non-technical: field devices need updates; a bad update can brick one; Helix keeps the old version bootable until the new one proves itself.	SHIPPING
S2	The problem	Single-slot update fails mid-flash = dead device + truck roll → sets up the A/B payoff.	SHIPPING
S3	How it works	The technical spine, animated: slot A live / slot B receives → update_engine writes B → AVB verify → dm-verity → reboot → auto-rollback on failure; then zoom out to the Go control plane (releases, groups, staged rollout) + the on-device agent + telemetry.	SHIPPING
S4	Why it’s a game-changer	3-5 plain differentiators: never-brick (hardware-backed rollback), cryptographic trust (verify key from server config only), self-hostable	SHIPPING

#	Section	Job (message intent — final copy is a later, operator-owned step)	Maturity shown
		control plane, delta updates, emergency recall.	
S5	Full feature showcase	Bento grid of every REAL capability grouped A-E (\$1.3 of research), each tile deep-links to a docs anchor.	SHIPPING + ROADMAP tiles marked
S6	Architecture	System diagram: device fleet ↔ Go control plane ↔ artifact store; the six ota-* bricks labelled.	SHIPPING
S7	For power users	Self-host, REST API + (roadmap) upload CLI, the reusable Go/Kotlin bricks, supported targets + roadmap, the OTA Manager operator console.	SHIPPING + ROADMAP marked
S8	CTA / Docs / Footer	Get-started, docs link, GitHub, license; the operator-locked “ Contact sales ” CTA → mailto:contact@hxota.com (NO pricing); footer “ <i>Made with ♥ by Helix Development team</i> ”.	—

1.2 Deferred deep routes (optional — add when content depth justifies, not blocking)

Route	Purpose	Rendering	Launch?
/features	Long-form expansion of S5 (SEO surface, keeps home lean)	prerendered (SSG)	deferred
/architecture			deferred

Route	Purpose	Rendering	Launch?
	Deep architecture + the ota-* bricks	prerendered (SSG)	
/docs	External link-out to the docs repo/site (not rebuilt here)	external	link only

1.3 Required machine / utility pages

sitemap.xml, robots.txt, a branded 404, and (recommended) a /license or footer license link (the bricks are Apache-2.0). These are part of the SEO surface (\$4), not narrative pages.

Page-count summary: 1 indexable HTML route at launch (single long-scroll home, 9 sections) + 2 optional deferred deep routes + 1 external docs link + the machine files (sitemap.xml, robots.txt, 404).

2. Information architecture + per-page semantic HTML5 content outline

The document outline is authored for SEO and accessibility from the first line: exactly one `<h1>` per route, a strict heading hierarchy (no skipped levels), landmark elements (`<header>/<nav>/<main>/<section>/<footer>`), and aria-labelledby binding each `<section>` to its heading so the accessibility tree and the crawl outline agree.

2.1 / — semantic outline (launch home)

```

<header>                                     ← site masthead (sticky)
  <nav aria-label="Primary">                 ← logo, section anchors,
theme + language switchers
<main>
  <h1> Ship firmware to your whole fleet. Never brick a device.
← S0, the ONE h1
  <section aria-labelledby="story">    <h2 id="story">...</h2>
← S1
  <section aria-labelledby="problem"> <h2 id="problem">...</h2>
← S2

```

```

    <section aria-labelledby="how">      <h2 id="how">How it works</
h2>← S3
      <h3> A/B partitions </h3> <h3> Verify + rollback </h3> <h3>
Control plane </h3>
    <section aria-labelledby="why">      <h2 id="why">...</h2>
← S4
    <section aria-labelledby="features"><h2 id="features">Features</
h2>← S5 (bento; each tile <article><h3>)
    <section aria-labelledby="arch">      <h2 id="arch">Architecture</
h2>← S6 (figure + <figcaption>)
    <section aria-labelledby="power">      <h2 id="power">For power
users</h2>← S7
    <section aria-labelledby="cta">      <h2 id="cta">Get started</
h2>← S8 (Contact-sales CTA)
<footer>                                ← Made with ♥ ...,
contact@hxota.com, GitHub, license

```

- **Diagrams** are <figure> + <figcaption> (the caption is crawlable, SEO-relevant text; the SVG carries role="img" + <title>/<desc>).
- **The hero WebGL canvas** is decorative (aria-hidden), with the real headline as live DOM text so the prerendered HTML contains the promise (SEO + no-WebGL fallback).
- **In-page anchors** (#how, #features, ...) are the deep-link + nav targets.

2.2 /features, /architecture (deferred)

Same discipline: one <h1> (the page topic), <h2> per capability group, <article> per feature, a BreadcrumbList (§4.4) linking back to /. / architecture carries the prerendered Mermaid → SVG lifecycle diagram as a captioned <figure>.

3. Responsive design

Layout approach: CSS **Grid** for page/section macro-layout (bento grid, architecture columns), **Flexbox** for component micro-layout (nav, cards, CTA rows), **fluid type** via the OpenDesign token ramp with clamp() between tier steps (never ad-hoc font sizes — DESIGN.md caps at three type sizes per screen). Container capped at the token -- container-max (1200px); gutters/section-rhythm come straight from the tokens (24/16/12 gutter, 80/48/32 section-Y). No fixed pixel layouts; every width is relative, every image max-width:100%, and any wide

element (architecture SVG, code block) scrolls inside its own overflow-x:auto container so the page body never scrolls sideways. (Research §5.)

3.1 Breakpoint × device-class matrix (the responsiveness to be PROVEN, §11.4.190(A))

Device class	Representative viewport(s)	Gutter	Section-Y	Hero treatment	Feature grid
Phone	360, 390	12px	32px	static poster PNG + CSS mesh (battery/perf); headline steps -- text-4x1 → -- text-2x1	1-col stack
Tablet	768, 834	16px	48px	lighter WebGL (reduced particles)	2-col
Laptop	1024, 1280	24px	80px	full OGL WebGL + kinetic headline	3-col
Desktop	1440	24px	80px	full WebGL	3-col
Large display	1920, 2560	24px (content capped 1200)	80px	full WebGL, content centered	3-col, generous margins

3.2 Engine matrix (the browsers to be PROVEN)

Chromium · Firefox · WebKit — the three engines that cover Windows/macOS/Linux + Android/iOS mobile. No vendor-only APIs; WebGL is feature-detected with a poster fallback; `color-mix()`/`oklab()` accent tokens ship a static-hex fallback layer for older engines. Proof runs as Playwright projects (§7).

Proof matrix (host-rendered): 5 device classes × 3 engines × {light, dark} = 30 base render combinations per key screen/section, plus the two first-class device-state snapshots **prefers-reduced-motion: reduce** (static composition) and **no-WebGL** (poster fallback). Interactive targets ≥ 44px; hover-reveal content has a tap/always-on fallback behind @media (hover: hover) and (pointer: fine).

4. SEO plan (§11.4.190(B))

The marketing site is the **only indexable surface** of the whole system (the two admin SPAs stay noindex, per firebase.json intent). SEO is therefore fully owned here.

4.1 Per-page <title> + meta description (proposed)

Route	<title> (≤ 60 chars)	<meta name="description"> (≤ 155 chars)
/	Helix OTA – Safe A/B Firmware Updates for Fleets	Ship firmware to your whole device fleet with native Android A/B updates, cryptographic verification, and automatic rollback — on a self-hostable control plane.
/features	Helix OTA Features – A/B, Rollout, Recall, Delta	Every Helix OTA capability: native Android A/B updates, staged rollouts, emergency recall, delta updates, telemetry, and a self- hostable Go control plane.
/architecture	Helix OTA Architecture – Fleet · Control Plane	How Helix OTA works end-to-end: device fleet, Go control plane, artifact store, and the

Route	<title> (≤ 60 chars)	<meta name="description"> (≤ 155 chars)
		reusable ota-* protocol bricks.

Each is unique, front-loads the primary keyword, and contains no pricing token (content lock #2). Final wording is operator-owned (§9-D); these are the SEO-shaped defaults.

4.2 Open Graph + Twitter cards (every route)

og:type=website, og:site_name=Helix OTA, per-route og:title/
og:description/ og:url (absolute, canonical domain), og:image = a
1200×630 branded social card (derived from assets/Logo.png + the
green gradient-mesh; committed asset, no external host), og:image:alt.
Twitter: twitter:card=summary_large_image, twitter:title/
twitter:description/twitter:image. Rendered into the **prerendered**
HTML <head> (not injected client-side) so crawlers see them without
executing JS.

4.3 Canonical + hreflang

Self-referential absolute <link rel="canonical"> per route (needs the
production domain, §9-C). hreflang is deferred until a 2nd locale ships
(English-only at launch); the runtime i18n switch is single-URL, so
multi-locale SEO is a documented revisit-point (research §4.2), not a
launch cost.

4.4 Structured data — JSON-LD (schema.org), anti-bluff constrained

Emit in prerendered <head>. **Proposed types:**

- **Organization** — publisher “Helix Development team”, logo
(absolute URL to the brand mark), sameAs → the GitHub/GitLab org
URLs, email contact@hxota.com.
- **WebSite** — site name + url. **No SearchAction** (there is no on-site
search → claiming one is a bluff).
- **SoftwareSourceCode** — for the OSS bricks: codeRepository → the
GitHub org, programmingLanguage Go + Kotlin, license Apache-2.0.
Honest and strong for an open-source product.

- **SoftwareApplication** — the product: applicationCategory “DeveloperApplication”, operatingSystem “Android (RK3588 / Orange Pi 5 Max)”. **No Offers/price** (content lock #2 — no pricing) and **no aggregateRating/review** (no real reviews exist — fabricating them is a \$11.4 bluff).
- **BreadcrumbList** — on the deep routes only.
- **FAQPage** — OPTIONAL, and only if a genuine FAQ section is authored with real answers; do not add an empty/fabricated FAQ purely for rich-results.
- **TechArticle** — OPTIONAL, for /architecture when it ships.

Anti-bluff rule for structured data: every JSON-LD field maps to a real, shipping fact; roadmap capabilities never appear as shipped featureList entries; the payload is validated (0 errors/warnings) by the §7 structured-data gate.

4.5 sitemap.xml + robots.txt

- `sitemap.xml` — lists / at launch (add /features, /architecture when they ship), generated at prerender time (not hand-maintained), `<lastmod>` from the build.
- `robots.txt` — Allow: /, Sitemap: absolute URL. The marketing site is indexable; the admin SPAs remain `noindex` on their own hosts.

4.6 Core Web Vitals + WCAG targets, and the Lighthouse score-floor

- **CWV targets:** LCP < 2.0 s · CLS < 0.02 · INP < 200 ms · TBT < 150 ms (throttled). Initial JS on / (excluding @defer-loaded WebGL/GSAP) < 150 KB gzip. (Research §2.8.)
 - **WCAG AA:** axe-core clean per route + the programmatic contrast oracle (§5.4 / §7) recomputing every accent-on-surface pair ≥ 4.5:1 text / 3.0:1 UI.
 - **Lighthouse score-floor (proposed, the §11.4.190(B) “defined score floor”):** SEO = 100 (a static marketing page has no excuse for less), Accessibility = 100, Performance ≥ 95, Best-Practices ≥ 95, structured-data validation = 0 errors. Every floor is a blocking CI gate (§7), not a hope.
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5. OpenDesign token system (§11.4.162)

5.1 What exists vs what the marketing site needs

The repo ships a real OpenDesign package at `design-systems/helix-ota/` (`od-design-system-project/v1`, id `helix-ota`, Enterprise) — but it is **dashboard-first with a BLUE accent**, and its own `DESIGN.md` says it is “not for marketing pages.” The marketing site therefore **consumes the vendored OpenDesign tokens and authors a Helix-green brand LAYER on top** (§11.4.74 extend-upstream, §11.4.162 no-ad-hoc-CSS) — it does not fork the palette and does not use the stock blue as-is.

How the tokens define the system (from the vendored package):

`tokens.css` is the source of truth (light base `:root` + three dark selectors: `@media (prefers-color-scheme: dark), [data-theme="dark"], .dark`); `tailwind-v4.css` maps every token custom-property into a Tailwind v4 `@theme --color-*/--spacing-*/--text-*/--radius-*/--shadow-*` entry, so Tailwind utilities resolve straight to tokens. Palette (light/dark): `--bg #FFFFFF/#020817`, `--surface-warm #F1F5F9/#1E293B`, `--fg #020817/#F8FAFC`, `--border #E2E8F0/#1E293B`; type ramp `12·14·16·20·24·32·48·64`; radius `sm/md/lg`; three elevation levels. (Read `design-systems/helix-ota/{DESIGN.md,tokens.css,tailwind-v4.css,manifest.json}`.)

5.2 The Helix-green brand layer (designed in research §3, contrast-verified in 01_ * §1)

Author one new file `src/design-tokens/brand-helix-green.css`, imported **after** `tokens.css`, that (a) overrides the accent slots to the lime FILL and (b) adds brand tokens — preserving every OpenDesign schema name. The key semantic inversion (why this is not a hue swap): the lime `#BBE639` is a **light** accent, so:

- `--accent` = lime fill, `--accent-on` = near-black-green (dark text on lime, $\approx 13:1$) — the inverse of the blue’s near-white on-color.
- Lime/mint **cannot** be accent TEXT on a light background (lime-on-white $\approx 1.45:1$, computed FAIL). Add `--accent-ink: #3f6212` (deep green, links/text, $7.08:1$ on white) and `--accent-strong: #4d7c0f` (large text/icons); mint `#B8ECD7` → `--accent-soft` (tints/badges/mesh stops).
- Dark theme: `--accent-ink #c9ed5b` ($14.10:1$), `--accent-strong mint` ($14.35:1$), optional brand-green-tinted near-black `--bg #071410`.

Every one of these contrast pairs was **COMPUTED** and **PASSES** its **AA bar** (research 01_* §1.2/§1.4, WCAG 2.1 relative-luminance script). The table is the design intent; the §7 contrast oracle recomputing every pair against the surface it actually lands on is the proof (§11.4.6 — no assumed AA).

5.3 Brand direction (unique, not templated)

Concept: “**The Living Helix / Green Signal**” — a double helix whose two strands ARE the A/B partitions; the update flows down one strand while the other stays lit and bootable, and snaps back on rollback. The motif is literally the product story. Backed by a green gradient-mesh (lime → mint → emerald → deep teal), mono checksum micro-details, an inverted bright-lime CTA on dark, and self-hosted display type (Space Grotesk display / Hanken Grotesk body / JetBrains Mono) — recognizably Helix, not a stock SaaS theme. (Research §6.)

5.4 UNCONFIRMED / what’s needed for OpenDesign completeness (§11.4.6)

- **UNCONFIRMED: the OpenDesign DAEMON/MCP plugin is NOT built or run** (operator-gated; docs/research/opendesign_daemon_setup_20260709/ = “nothing built or run”). “Heavy OpenDesign use” is therefore defined as **CONSUME the vendored tokens + author the §5.2 brand-green layer** — not running a live daemon. **Needed:** operator confirmation that tokens-only is the accepted meaning for launch (K13 decision (b), §9-C).
 - **UNCONFIRMED: the Helix-green brand layer is not yet part of the OpenDesign package.** It is fully designed + contrast-computed in the research, but design-systems/helix-ota/ still ships only the blue dashboard tokens. **Needed:** decide whether the green layer is 1. authored in the website repo only (fastest; the site is a separate consumer, §11.4.28) or (ii) promoted upstream into the OpenDesign package as a marketing variant (reusable, more work). Proposal: **(i) for launch**, revisit (ii) later.
 - **UNCONFIRMED: dark-surface sub-decision** — green-tint #071410 vs shipped slate #020817 (research §3.5). Both pass AA; a brand-taste call (K13, §9-C).
 - **UNCONFIRMED: production domain** for canonical/OG/sitemap absolute URLs. The contact email is @hxota.com, but the canonical web domain is not stated — must be confirmed, never guessed (K13, §9-C).
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6. Tech-stack recommendation

Operator mandate context (research §authority): the operator has mandated **Angular** + “the most advanced visual tech that FITS OpenDesign.” §11.4.190 additionally wants **static-first** output for Core Web Vitals + SEO. The two are reconcilable: Angular 22 with **full prerender (SSG)** emits static HTML per route — static-first output under the mandated framework. Three options, honestly compared:

Option	What	Pros	Cons	§11.4.1
A. Angular 22 SSR+prerender (SSG) + Tailwind v4 on tokens (RECOMMENDED)	Standalone + signals, --ssr then prerender every route, provideClientHydration() + incremental hydration; Tailwind v4 CSS-first wired to the OpenDesign @theme tokens; GSAP + OGL deferred.	Honors the Angular mandate; first-class SSR/ prerender; strong DI for Theme/i18n; batteries-included test surface; prerendered routes = full SEO + fast FCP; Tailwind-on-tokens satisfies §11.4.162 with zero palette fork.	Heavier framework than a pure static generator; hydration must be disciplined (browser-only WebGL guards); needs a build-time route list (trivial here).	Strong prerender static output, budget enforced tokens native.
B. Astro + Tailwind v4 on tokens	Islands architecture, ships near-zero JS by default, static-first.	Best raw CWV/SEO ceiling (least JS); simplest static hosting.	Contradicts the Angular mandate; re-implements the ThemeService/i18n outside Angular; team/consistency drift from the	Strong CWV, but fails the mandate unless operator lifts it.

Option	What	Pros	Cons	§11.4.1
C. Plain semantic HTML + CSS (+ a little vanilla JS)	Hand-authored static pages, tokens via plain CSS custom properties.	Absolute-minimum bytes; no framework; trivial hosting; maximal CWV.	two Angular-adjacent SPAs (they are React, not Angular — but Astro adds a third paradigm). No component system for the bento/diagram/animation reuse; i18n + theme + the “most advanced visual tech” mandate become hand-rolled; scales poorly if / features+ / architecture grow; §11.4.162 “no ad-hoc CSS” harder to honor without a system.	CWV excellence weak c “advan visual OpenD compo manda

Recommendation: Option A (Angular 22 SSR+SSG + Tailwind v4 on OpenDesign tokens). It is the only option that honors the operator’s Angular mandate AND produces static-first prerendered output for §11.4.190’s CWV/SEO requirement AND consumes the OpenDesign tokens natively via Tailwind’s @theme. The full stack (verified current in research 01_* §2): Angular 22 · Tailwind v4 (@tailwindcss/postcss) · GSAP + ScrollTrigger (scroll story, free for commercial use) · OGL (~10-20 KB hero WebGL, not Three.js) · @jsverse/transloco v8 (signals, instant-switch i18n) · self-hosted Fontsource fonts (OFL 1.1) · lucide-angular icons · pnpm · Playwright/axe/Lighthouse CI for proof.

Whether to keep the Angular mandate or accept Astro (Option B) for a lighter build is itself an operator call (K13, §9-C) — flagged, not silently taken.

7. Anti-bluff proof plan (§11.4.190(E) / §11.4.170)

Value/token-equality unit tests are **FORBIDDEN as the sole UI proof** (§11.4.170). Every UI/SEO/quality claim is proven by captured evidence under `docs/qa/<run-id>/` (§11.4.83), reproducible inside the containerized build path (§11.4.173, rootless podman §11.4.161). (Research §7.)

§11.4.190 claim	How it is CAPTURED as evidence	Gate
(A) Responsiveness	Playwright host-renders every section/component across the §3 matrix (5 device classes × {chromium, firefox, webkit} × {light, dark}) + the reduced-motion & no-WebGL states → committed PNGs.	CM-WEBSITE-RESPONSIVE-PROVEN-class
Layout correctness	OCR / vision layout oracle reads rendered headlines/labels/control bounds and asserts NO overlap / label-over-label / clipping / off-screen / collapsed-or-giant widget.	same
Visual regression	Golden image-diff (toHaveScreenshot/pixelmatch) with a self-validated golden-good + golden-bad pair per component (§11.4.107(10)) — a deliberately broken fixture MUST fail, or the analyzer is itself a bluff.	same
(B) SEO	Lighthouse CI vs the §4.6 floors (SEO=100, A11y=100,	CM-WEBSITE-SEO-OPTIMIZED-class

§11.4.190 claim	How it is CAPTURED as evidence	Gate
WCAG AA	Perf≥95) + structured-data validation (0 errors) + presence checks for per-route title/meta/OG/canonical/sitemap/robots. axe-core/playwright per route + the programmatically contrast oracle recomputing every accent-on-surface pair, FAIL below 4.5 text / 3.0 UI (the machine proof the §5.2 table defers to).	a11y gate
(C) OpenDesign uniqueness	Token-provenance check — every color/space/type value traces to a token (no raw hex outside :root), and the brand layer extends (not forks) the vendored package.	CM-WEBSITE-OPENDESIGN-UNIQUE-class
(D) Enterprise visual quality (light+dark)	The §11.4.170 device-independent host-rendered pixel proof per screen × state × {light, dark} — the golden set above IS this proof.	§11.4.170
Content locks (§1.4)	OCR reads exact contact@hxota.com in S8 + footer (DOM mailto: href); OCR scans every route for price/currency/“plan”/“tier”/“\$”/“€” → FAIL on any hit + assert one “Contact sales” CTA; footer renders the Heart SVG (not the word “love”), AA in light+dark, accessible name “Made with love ...”.	content-lock gate
Behavior	Theme-switch (flip data-theme, persists, no FOUC on prerendered load) + language-switch (DOM text	e2e

§11.4.190 claim	How it is CAPTURED as evidence	Gate
	changes, longest-string layout intact, no untranslated-key leak) + SSR/hydration (zero mismatch console errors, prerendered HTML contains real headline text).	
Final human gate	§11.4.185 manual QA-team confirmation is the LAST step (automation is necessary, not sufficient); manual QA never substitutes the automated proof, and the automated proof never substitutes manual QA.	§11.4.185

Honest boundary (§11.4.6): the site is not scaffolded, so these are DESIGN CAPTURES — no screenshot/OCR/Lighthouse evidence is producible until the scaffold exists, and none is claimed here. They become live gates the moment P4.5 runs.

8. Phased build plan (P4 sub-phases → T4 / feature/website / submodules/website)

Maps the production plan's **P4** (effort **L**) into ordered sub-phases. All work lands on the canonical branch **feature/website** used identically on the parent repo AND the new **submodules/website** (§11.4.181/§11.4.191), trunk-merged regularly (§11.4.188), merged to main **ONLY** after operator approval + §11.4.185 manual QA (§11.4.167(I)). Effort = T-shirt only (§11.4.172 — no false calendar precision until velocity is measured).

Sub-phase	Work	Gates produced	Effort
P4.0 Decide + bootstrap	Resolve K13 (§9); create submodules/ website (own .git §11.4.179, install_upstreams	token-drift fingerprint	S

Sub-phase	Work	Gates produced	Effort
P4.1 Tokens + brand layer	§11.4.36, .gitignore §11.4.30, README §11.4.44, helix-deps.yaml §11.4.31); vendor the OpenDesign tokens + scripts/ sync_design_tokens.sh sha256 fingerprint (§11.4.86); scaffold Angular 22 SSR + Tailwind v4. Author brand-helix-green.css + brand-theme.css (§5.2); ThemeService (signals, 3-toggle, no-FOUC inline script); wire the contrast oracle.	contrast oracle	S-M
P4.2 Layout shell + components	Container/type/grid primitives on tokens; nav, footer (heart + contact@hxota.com, content lock #3), theme + language switchers, contact-sales-cta, SVG diagram primitives, WebGL-hero shell (deferred, browser-guarded).	component goldens (good/bad)	M
P4.3 Pages / sections	S0-S8 scroll home; OGL hero + GSAP ScrollTrigger A/B-rollback & architecture diagrams; bento feature grid (SHIPPING/ROADMAP marked); responsive per §3.	per-section goldens across matrix	M-L

Sub-phase	Work	Gates produced	Effort
P4.4 SEO + i18n	Per-route title/meta/OG/canonical, JSON-LD (§4.4), prerender + sitemap.xml/robots.txt; Translocoben.json (switcher wired, English-only); TransferState for SSR locale/theme.	Lighthouse SEO + structured-data gate	M
P4.5 Proof + evidence	Playwright matrix (5×3×{light,dark} + reduced-motion + no-WebGL), golden diff + OCR oracle, axe, Lighthouse CI, content-lock OCR gate; capture all evidence under docs/qa/<run-id>/.	CM-WEBSITE-RESPONSIVE-PROVEN/-SEO-OPTIMIZED/-OPENDESIGN-UNIQUE	M
P4.6 Containerized build + deploy + QA handoff	Release build + §7 render/proof inside the containers-submodule build image (rootless podman §11.4.161/§11.4.173) on the remote build host; deploy to the chosen target (§9-B); hand off to QA for §11.4.185 confirmation.	build-in-container + deploy	S-M

Total: effort L (consistent with PRODUCTION_READINESS_PLAN.md §5 P4). Corresponds to plan workable-items #33-#36 (K13 resolve → scaffold → content → §11.4.190 proofs).

9. OPEN DECISIONS for the operator (K13 — precise questions)

These BLOCK P4 (production plan K13). Each is a decision, not a recommendation to silently take (§11.4.66). Recommendations are given; the operator chooses.

(A) Repository location & remote. 1. New submodule remote org + name? **Recommend** `git@github.com:HelixDevelopment/helix_ota_website.git` (OTA-domain convention, project-prefixed + greppable, snake_case §11.4.29); submodule PATH stays `submodules/website`. Alternatives: name `website` (shorter, less greppable) or org `vasic-digital`. 2. Confirm the four upstreams for the `upstreams/recipes` (GitHub primary + GitLab + GitFlic + GitVerse, §2.1).

(B) Build & deploy target + tokens/secrets. 3. Where is the built static site hosted? Options: add a third **Firebase Hosting** website target (alongside `dashboard/ota-manager`, but this one **indexable** — the admin SPAs stay `noindex`); or a static CDN / Cloudflare Pages / GitHub Pages / own server. **Recommend** a static/CDN target (SSG output). Confirm the **production domain** for canonical/OG/sitemap absolute URLs — the contact email is `@hxota.com` but the web domain is **UNCONFIRMED**: and must not be guessed. 4. Containerized production build sign-off (§11.4.173): **Recommend yes** — release build + §7 proof run in the `containers-submodule` build image (rootless podman) on the designated remote host; confirm the host + that the image exposes Node 24 / pnpm (extend upstream §11.4.74 if missing). 5. Secrets/tokens: does deploy require any secret (e.g. a hosting deploy token)? Is web **analytics** wanted at all, and if so which privacy-respecting option (or none)? Any secret is gitignored (§11.4.10/§11.4.30), never committed.

(C) Brand / design direction. 6. Confirm **OpenDesign-tokens-only** for launch (the daemon/MCP is not built) — **Recommend yes**; the §5.2 brand layer is what the daemon would emit. 7. Confirm the **Helix-green brand direction** + the “Living Helix” concept as the unique visual identity (vs any alternative the operator has in mind). 8. Dark surface: **green-tint #071410** vs shipped **slate #020817** (§5.4 / research §3.5) — both pass AA; a taste call. **Recommend green-tint** for the marketing surface. 9. Where does the Helix-green layer live: **website repo only** (recommend, launch) vs promoted into the OpenDesign package as a marketing variant (later)?

(D) Content ownership. 10. Who authors/approves the FINAL marketing copy? This proposal fixes intent + SEO-shaped defaults; the shipping words are operator/marketing-owned. 11. Who provides/approves the OG social-card image (1200×630, derived from `assets/Logo.png`)? 12. Confirm the SHIPPING-vs-ROADMAP maturity markings (§1, research §1.3) are acceptable as the honest-content policy, and who owns future locale files (`de.json`, ...) when i18n expands.

10. Honest boundary (§11.4.6)

- **This is a design proposal, not a build.** No repo was created, no code written, no git run — the only artifact is this document. The scaffold executes only after the operator approves this design (brainstorming HARD-GATE) + resolves the §9 (K13) decisions.
 - **It consolidates prior research** (`docs/research/website/00_* + 01_*`, Rev 2) into the requested proposal shape; those artifacts hold the fuller CSS/contrast/scaffold detail and are the cited source for §5-§7.
 - **OpenDesign specifics are partially UNCONFIRMED** (§5.4): the daemon/MCP is not built, the Helix-green marketing layer is designed-but-not-yet-in-the-package, the dark-surface and domain choices are open — each flagged, none guessed.
 - **Contrast numbers are computed static WCAG 2.1 ratios** (research `01_* §1`, PASS) — the token-level intent; they do NOT replace the §7 rendered-pixel contrast oracle that must recompute every pair against the actual surface (gradient-mesh stops, overlaps).
 - **Every recommendation carries its alternative**; nothing load-bearing is silently decided.
-

Sources verified (cited this session)

- Repo assets: `design-systems/helix-ota/`
`{DESIGN.md,tokens.css,tailwind-v4.css,manifest.json}`
(OpenDesign `od-design-system-project/v1`, blue dashboard tokens, “not for marketing pages”); `assets/Logo.png` (present); `firebase.json` (two admin-SPA hosting targets only); `.gitmodules` (no website submodule).
- `docs/research/website/00_WEBSITE_DESIGN_AND_BUILD_PLAN.md` (Rev 2) — narrative/IA, tech stack, Helix-green brand layer, responsive, visual identity, proof strategy, scaffold plan, operator decisions.

- docs/research/website/01_SCAFFOLD_READINESS_AND_VERIFICATION.md (Rev 2) — computed WCAG contrast PASS, Angular 22 / Tailwind v4 / Transloco / font-license verification against fetched official sources (2026-07-10).
- docs/planning/PRODUCTION_READINESS_PLAN.md (Rev 1) — §2.2 (website absent), §5 P4 (effort L), K13 (website decisions), workable-items #33-#36.
- Constitution §11.4.190 (website engineering-quality), §11.4.162 (OpenDesign), §11.4.170 (host-rendered pixel proof), §11.4.167/.181/.185/.188/.191 (feature-stream lifecycle, branch binding, manual-QA gate) — constitution/Constitution.md / project CLAUDE.md.